Foreign Direct Investment and Business Investment in Canada

Porokmi Sambahamphe
M.A. Economics student, Fordham University
psambahamphe@fordham.edu
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Abstract

Higher foreign direct investment (FDI) inflows are associated with lower business-investment changes in Canada over 1981–2018 when the data are measured in Canadian dollars, but the association is sensitive to how trends and expenditure are modelled. Using annual data from Statistics Canada and the World Bank, I estimate a change equation that relates business investment to expenditure, capacity slack, and current and lagged FDI. The sum of the current and lagged FDI coefficients is −0.130 (95 percent interval −0.188 to −0.073). It remains negative with expenditure excluding business investment, with Statistics Canada's own FDI series and after omitting any single year, but it falls to −0.050 and is not statistically significant when a linear trend is added with total expenditure as the control. An earlier US-dollar version of this analysis found positive but insignificant coefficients; an exact algebraic bridge attributes most of the difference to exchange-rate revaluation that enters annual changes measured in US dollars. The estimates do not establish that foreign inflows displace domestic investment. FDI measures financial inflows, which can include acquisitions of existing assets, rather than additions to productive capacity alone.

1 Introduction

Foreign direct investment can finance new plants, equipment, and technology, but a recorded inflow need not create an equivalent amount of fixed capital. Foreign investors may acquire an existing firm or provide financing that substitutes for funds available elsewhere. Domestic firms may respond by expanding production, supplying foreign affiliates, or reducing investment in markets where competition increases. Whether FDI adds to business investment is therefore an empirical question about the response of capital formation, not simply the volume of foreign funds received.

The question matters for Canadian investment policy. Under the Investment Canada Act, acquisitions subject to net-benefit review are assessed for their likely benefit to Canada, including effects on economic activity, employment and productivity (Investment Canada Act, ss. 20–21), and weak business investment has been a recurring policy concern (Rogers 2024). The Canadian data also show why a recorded inflow and an addition to capital differ. FDI inflows peaked at about C$129 billion in 2007, equivalent to about 42 percent of business investment that year, in a period that Statistics Canada (2008) attributes mainly to cross-border mergers and acquisitions. They then fell by more than four-fifths within two years, while business investment moved far less (Figure 1).

I examine this question for Canada using annual observations from 1981 to 2018, with an extension through 2024. The dependent variable is business gross fixed capital formation, and the explanatory variables are gross national expenditure, estimated capacity slack, and FDI net inflows. The investment series covers the business sector as a whole, including foreign-owned

firms. It therefore measures investment undertaken in Canada rather than investment by domestic owners alone.

The analysis revisits my 2019 undergraduate study, which estimated a levels regression in US dollars and interpreted a positive FDI coefficient as evidence of complementarity. An earlier version of this paper found positive but insignificant US-dollar coefficients alongside a negative Canadian-dollar association. This revision measures all series in Canadian dollars, the currency in which the investment is recorded and undertaken. Converting at a time-varying exchange rate adds a revaluation component to annual changes: in 2015, business investment fell by about C$15 billion, but its US-dollar value fell by US$62 billion, about US$51 billion of which reflects the depreciation of the Canadian dollar. The revision also allows FDI to enter with a one-year lag and tests the timing restrictions that earlier specifications imposed.

In Canadian dollars, higher FDI inflows are associated with lower investment changes over the current and following year, mainly through the lagged term. The association is negative in every Canadian-dollar specification examined and significant in most, but it is not significant when a linear trend is added with total expenditure as the control. An exact bridge between the currency specifications attributes most of the difference from the US-dollar estimates to revaluation. These comparisons show how much the finding depends on the specification; they do not identify an exogenous change in foreign investment. A companion Extended Working Paper (Sambahamphe 2026) reports the full set of checks.

2 Background and investment model

Agosin and Mayer (2000) examine whether FDI crowds domestic investment in or out in developing economies over 1970–1996. Their regional estimates show crowding in in Asia and crowding out in Latin America. Borensztein, De Gregorio, and Lee (1995), using data for 69 developing economies, report a more-than-proportional relationship between FDI and total investment. These findings suggest that the investment response depends on the receiving economy and the empirical specification.

Van Loo (1977) examines the same question for Canada and provides the reference for the earlier version of this study. Van Loo's reported FDI coefficient of about 1.40 exceeds one, whereas the estimates here are small and, in Canadian dollars, negative; differences in sample, measurement, and model specification prevent attributing this gap to a historical change in the investment response.

$$I_t = \alpha + \beta Y_t + \gamma U_t + \delta FDI_t + \varepsilon_t \quad (1)$$

I denotes business gross fixed capital formation, Y gross national expenditure, and U estimated capacity slack. Higher expenditure is expected to increase investment, while greater unused capacity is expected to reduce it. The expected signs are therefore $\beta > 0$ and $\gamma < 0$. The sign of δ is an empirical question. I also test whether FDI has a stronger investment association when capacity is available: near full capacity, additional demand may raise prices rather than expand production. Since slack measures output below potential, this hypothesis predicts that the FDI association becomes more positive as slack increases.

Two measurement issues affect interpretation. Expenditure includes investment: because Y = I + N, holding Y fixed while I varies implies an offsetting movement in other expenditure N, so I

also report specifications that control for N = Y − I. FDI is a financial-flow measure and need not correspond to new fixed capital formation in the same year. As Agosin and Mayer (2000) note, acquisitions transfer ownership, while foreign firms can also finance investment domestically. A coefficient below one does not by itself establish displacement of domestic firms.

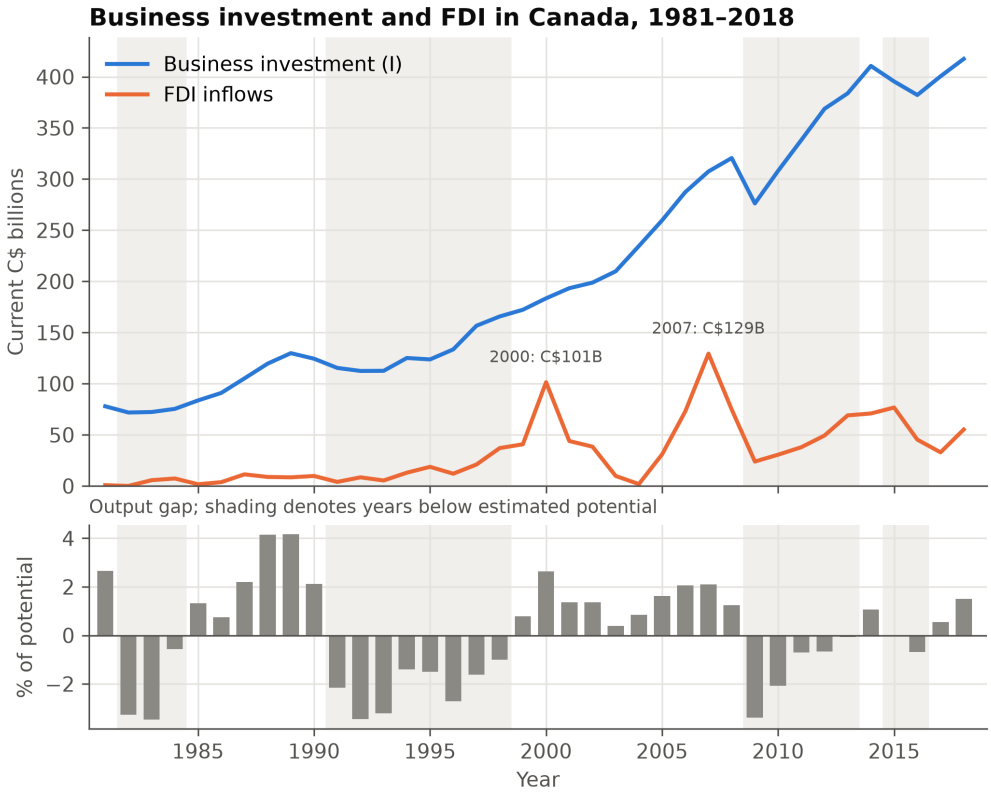
3 Data and methods

3.1 Data

The main sample contains 38 annual observations for 1981–2018, with an extension through 2024. Statistics Canada Table 36-10-0222-01 supplies business gross fixed capital formation, nominal GDP, and real GDP. World Development Indicators supplies gross national expenditure (NE.DAB.TOTL.CD), FDI net inflows (BX.KLT.DINV.CD.WD), and the annual average exchange rate (PA.NUS.FCRF).

Monetary variables are measured in current-price billions of Canadian dollars. World Bank expenditure and FDI are converted to Canadian dollars at the annual average exchange rate; for expenditure, this recovers the World Bank's own local-currency series to within 0.14 percent. The converted FDI series closely tracks Statistics Canada's direct investment liabilities (correlation 0.99986 over 1981–2018). No estimation variables are missing in the main sample. The earlier undergraduate paper reports 36 observations; the reconstructed sample contains 38. Differencing and the one-year FDI lag leave 37 observations.

Figure 1 Business investment and FDI inflows



Notes: Current C$ billions. Shading marks years below estimated potential output. Sources: Statistics Canada Table 36-10-0222-01; World Bank WDI; author's calculations.

3.2 Capacity slack

I estimate potential output by applying an HP filter to log real GDP over the available full series, with a smoothing parameter of 100. The log cycle is converted to a proportional gap and then valued at the current GDP price scale. In equation (2), τ denotes the HP trend in log real GDP.

$$c_t = ln(GDP_{t,real}) - \tau_t \quad (2)$$

$$g_t = exp(c_t) - 1 \quad (3)$$

$$U_t = GDP_{t,nominal} \times [\frac{1}{1+g_t} - 1] \quad (4)$$

Positive U indicates output below estimated potential. This measure differs from the earlier paper's stated definition of potential output minus gross national expenditure. Although the HP cycle removes an estimated trend, U is expressed on a changing nominal GDP scale, so its stationarity must be tested. Using the full series also makes the estimate sensitive to later observations and data revisions.

3.3 Estimation

The models are estimated in Python using statsmodels. Augmented Dickey–Fuller tests, with lag length selected by AIC, fail to reject a unit root in levels for Canadian-dollar investment and expenditure and reject it in first differences (Table 1). Slack rejects a unit root with or without a trend. FDI rejects a unit root only around a linear trend, not around a constant, and rises by about C$1.8 billion per year.

Engle–Granger tests of investment and expenditure do not reject the null of no cointegration in Canadian dollars under the default lag search (p = 0.382), but do under a capped BIC search (p = 0.023). Because the long-run evidence depends on the lag-selection setting, the baseline is a change equation without an error-correction term:

$$\Delta I_t = \kappa + \theta \Delta X_t + \gamma U_t + \delta_0 FDI_t + \delta_1 FDI_{t-1} + e_t \quad (5)$$

X is gross national expenditure Y or, as a check, expenditure net of business investment, N = Y − I. Equation (5) nests the two restrictions used in earlier specifications: FDI in levels ($\delta_1 = 0$) and FDI in first differences ($\delta_0 + \delta_1 = 0$). The main quantity of interest is the coefficient sum $\delta_0 + \delta_1$. For a one-year C$1 billion increase in FDI, it describes the combined association with investment changes in that year and the next, holding expenditure changes and slack fixed. Standard errors are Newey–West HAC with three lags. The baseline omits a linear trend as a presentation choice; Table 2 reports the trend specification alongside it, and that comparison is not evidence that either specification is statistically superior.

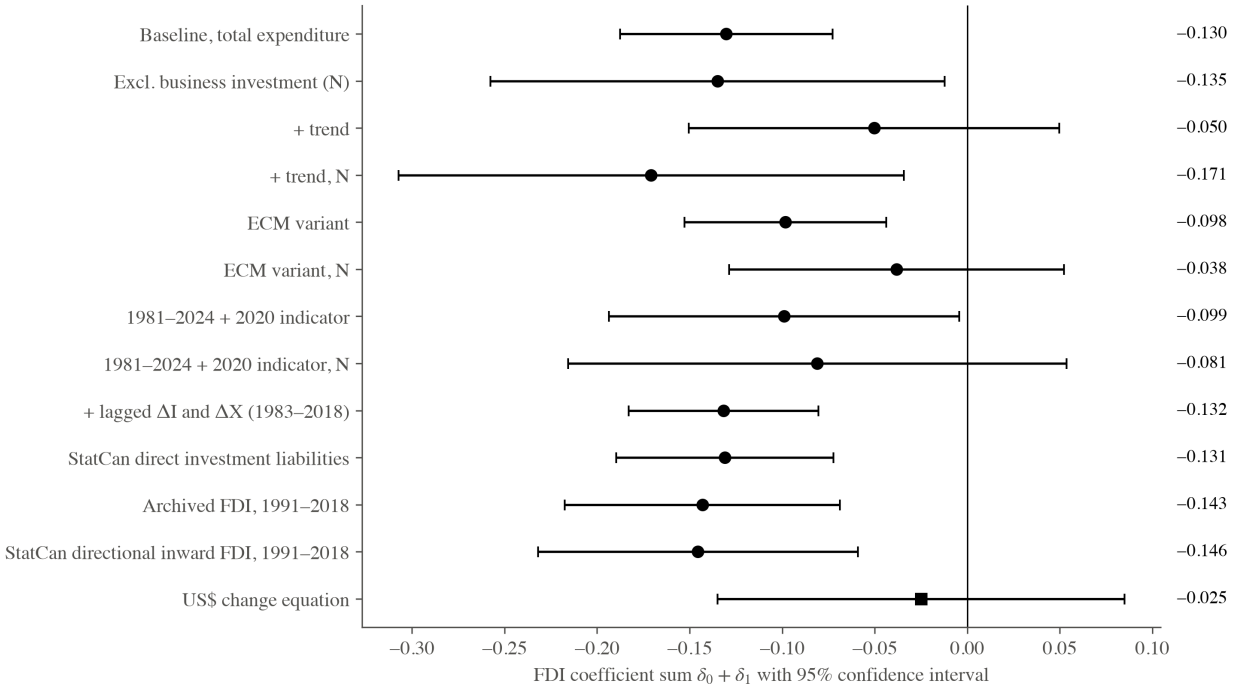
To explain why the currency of measurement matters, I translate the US-dollar equation exactly into Canadian dollars and then remove, one at a time, the weighting, the constant, the revaluation terms in differenced variables, the exchange-rate adjustment of lagged FDI, and the exchange-rate regressor. Each adjustment's contribution is averaged over all 120 orders in which the five adjustments can be applied. The revaluation terms arise because, for any monetary series Z and exchange rate e (Canadian dollars per US dollar),

$$\Delta Z_t^{USD} = \Delta Z_t^{CAD}/e_t + Z_{t-1}^{CAD}(1/e_t - 1/e_{t-1}) \quad (6)$$

The second term changes the US-dollar value of last year's level without any change in spending.

4 Results

Figure 2 FDI coefficient sum across specifications



Notes: Points show δ₀ + δ₁ with 95% confidence intervals from HAC standard errors (three lags). Circles: Canadian-dollar specifications; square: US-dollar change equation. Statistics Canada rows use current releases. Source: Author's calculations.

Table 1. Augmented Dickey–Fuller tests, C$, 1981–2018

Variable	p (level, constant + trend)	p (level, constant only)	p (first difference)
I	0.543	0.992	<0.001
Y	0.970	1.000	<0.001
U	0.010	0.004	<0.001
FDI	<0.001	0.849	<0.001

Null: unit root. Lag length by AIC. Differences include a constant.

Table 2. Baseline Canadian-dollar estimates

	(1) Total expenditure	(2) Excl. investment	(3) (1) + trend	(4) (2) + trend
ΔX	0.431	0.417	0.491	0.365
	(0.053)	(0.155)	(0.032)	(0.195)
U	-0.002	-0.123	0.064	-0.157
	(0.061)	(0.113)	(0.060)	(0.111)
FDI (δ_0)	-0.015	0.063	0.014	0.048
	(0.057)	(0.087)	(0.065)	(0.088)
FDI_{-1} (δ_1)	-0.116	-0.198	-0.064	-0.219
	(0.068)	(0.103)	(0.066)	(0.097)
Trend			-0.326	0.173
			(0.199)	(0.371)
Intercept	-9.032	-4.586	-8.345	-4.544
	(2.263)	(5.592)	(2.511)	(5.714)
$\delta_0 + \delta_1$	-0.130	-0.135	-0.050	-0.171
95% interval	[-0.188, -0.073]	[-0.258, -0.012]	[-0.150, 0.050]	[-0.307, -0.034]
p, $\delta_0 + \delta_1 = 0$	<0.001	0.031	0.324	0.014
p, $\delta_1 = 0$	0.089	0.055	0.329	0.024
Breusch–Godfrey p	0.223	0.274	0.289	0.295
N	37	37	37	37

Dependent variable ΔI, C$ billions, 1982–2018. X = Y in (1), (3); X = N = Y − I in (2), (4). HAC (3 lags) SEs in parentheses.

In the baseline (Table 2, column 1), the contemporaneous FDI coefficient is −0.015 and the lagged coefficient −0.116. Their sum is −0.130, with a 95 percent interval of [−0.188, −0.073]. For a one-year C$1 billion increase in inflows, annual investment changes are about C$0.13 billion lower in total over that year and the next, conditional on expenditure and slack. With expenditure excluding business investment (column 2), the sum is −0.135 (p = 0.031).

Evidence about timing is weaker than evidence about the sum. The data reject the differences restriction (p < 0.001), so entering FDI as an annual change is not supported, but they do not reject the levels restriction at 5 percent (p = 0.089). The baseline residuals show no evidence of serial correlation at two lags (Breusch–Godfrey p = 0.223).

Adding a linear trend (columns 3 and 4) matters when total expenditure is the control: the sum falls to −0.050, with an interval of [−0.150, 0.050]. With expenditure excluding investment it becomes −0.171 (p = 0.014). The negative association is therefore sensitive to how trends and expenditure are modelled. The expenditure-change coefficient is 0.431 (standard error 0.053), consistent with investment moving with demand, although the accounting overlap prevents a structural interpretation. The slack coefficient is close to zero (−0.002).

5 Robustness and limitations

Table 3. Sensitivity of the FDI coefficient sum

Specification	$\delta_0 + \delta_1$	95% interval	p	BG p	N
Baseline, total expenditure	-0.130	[-0.188, -0.073]	<0.001	0.223	37
Excl. business investment (N)	-0.135	[-0.258, -0.012]	0.031	0.274	37
+ trend	-0.050	[-0.150, 0.050]	0.324	0.289	37
+ trend, N	-0.171	[-0.307, -0.034]	0.014	0.295	37
ECM variant	-0.098	[-0.153, -0.044]	<0.001	0.149	37
ECM variant, N	-0.038	[-0.129, 0.052]	0.407	0.018	37
1981–2024 + 2020 indicator	-0.099	[-0.194, -0.004]	0.040	0.367	43
1981–2024 + 2020 indicator, N	-0.081	[-0.216, 0.054]	0.238	0.567	43
+ lagged ΔI and ΔX (1983–2018)	-0.132	[-0.183, -0.080]	<0.001	0.001	36
StatCan direct investment liabilities	-0.131	[-0.190, -0.072]	<0.001	0.222	37
Archived FDI, 1991–2018	-0.143	[-0.217, -0.069]	<0.001	0.377	28
StatCan directional inward FDI, 1991–2018	-0.146	[-0.232, -0.059]	<0.001	0.287	28
US$ change equation	-0.025	[-0.135, 0.085]	0.655	0.224	37

C$ unless stated; HAC (3 lags). BG: two-lag Breusch–Godfrey p. StatCan rows use current releases.

Table 3 collects the coefficient sum across alternative specifications. It is negative in every Canadian-dollar row and significant at 5 percent in all but three: the trend specification with total expenditure, and the error-correction and 1981–2024 variants with expenditure excluding investment. Replacing the World Bank FDI series with Statistics Canada's direct investment liabilities leaves the sum at −0.131; its directional inward-FDI series gives −0.146 over 1991–2018, against −0.143 for the archived series on the same years. Omitting any single year gives sums between −0.155 and −0.095.

The US-dollar change equation gives a sum of −0.025 ($p = 0.655$). The exact bridge moves the estimate from −0.025 to −0.130. Replacing revalued changes with Canadian-dollar changes contributes −0.096 of that −0.105 movement on average, and between −0.084 and −0.108 in every order; the other adjustments contribute between −0.017 and +0.020 on average. This is an accounting attribution along specified transformations, not a causal explanation. It shows that currency conversion is a specification change rather than a simple rescaling.

When current and lagged FDI are interacted with percent slack, the two interaction terms are jointly significant ($p < 0.001$) but have opposite signs, and their sum, which measures whether slack changes the combined association, is −0.060 ($p = 0.116$). Exploratory specifications adding indicators for 2001 and 2009 retain a negative baseline FDI coefficient sum but weaken the evidence for capacity interactions: the joint interaction p-value rises to 0.110 with total expenditure and 0.035 with expenditure excluding business investment (Sambahamphe 2026, Appendix D). The data therefore provide no evidence that slack strengthens the FDI association.

Several limitations remain. The specifications were developed during revision, after earlier results had been seen, and the sample is small for the number of modelling decisions. Expenditure contains investment, the investment series includes foreign-owned firms, and the

HP-based slack measure is a retrospective estimate that depends on the data vintage. The variants that add lagged investment changes, or combine error correction with expenditure excluding investment, show residual serial correlation. These features prevent the estimates from identifying displacement of domestic investment.

6 Conclusion

Measured in Canadian dollars, higher FDI inflows are associated with lower business-investment changes in Canada over 1981–2018, with a baseline coefficient sum of −0.130 concentrated in the following year. The association survives changes in the FDI source, the omission of single years and additional short-run dynamics, but it is sensitive to how trends and expenditure are modelled: with a linear trend and total expenditure as the control, it is −0.050 and not significant. The trend-free baseline is a presentation choice, not a specification the data show to be superior. The earlier positive US-dollar association does not survive, and most of the difference between the currency specifications reflects exchange-rate revaluation.

These results caution against treating aggregate foreign inflows as a direct measure of new capital formation. A negative association is consistent both with acquisition-driven inflows followed by consolidation of capital spending and with inflows that peak late in investment upswings, as in 2000 and 2007; the aggregate estimates do not separate those channels. For policy, the relevant distinction is between inflows that expand productive capacity and transactions that mainly change ownership or financing. Investment data by ownership and FDI entry mode would help determine which mechanism the negative association reflects.

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